DATABASE LAB PROJECT

A Relational Framework for Intelligent Job–Applicant Matching & Recruitment Lifecycle Management

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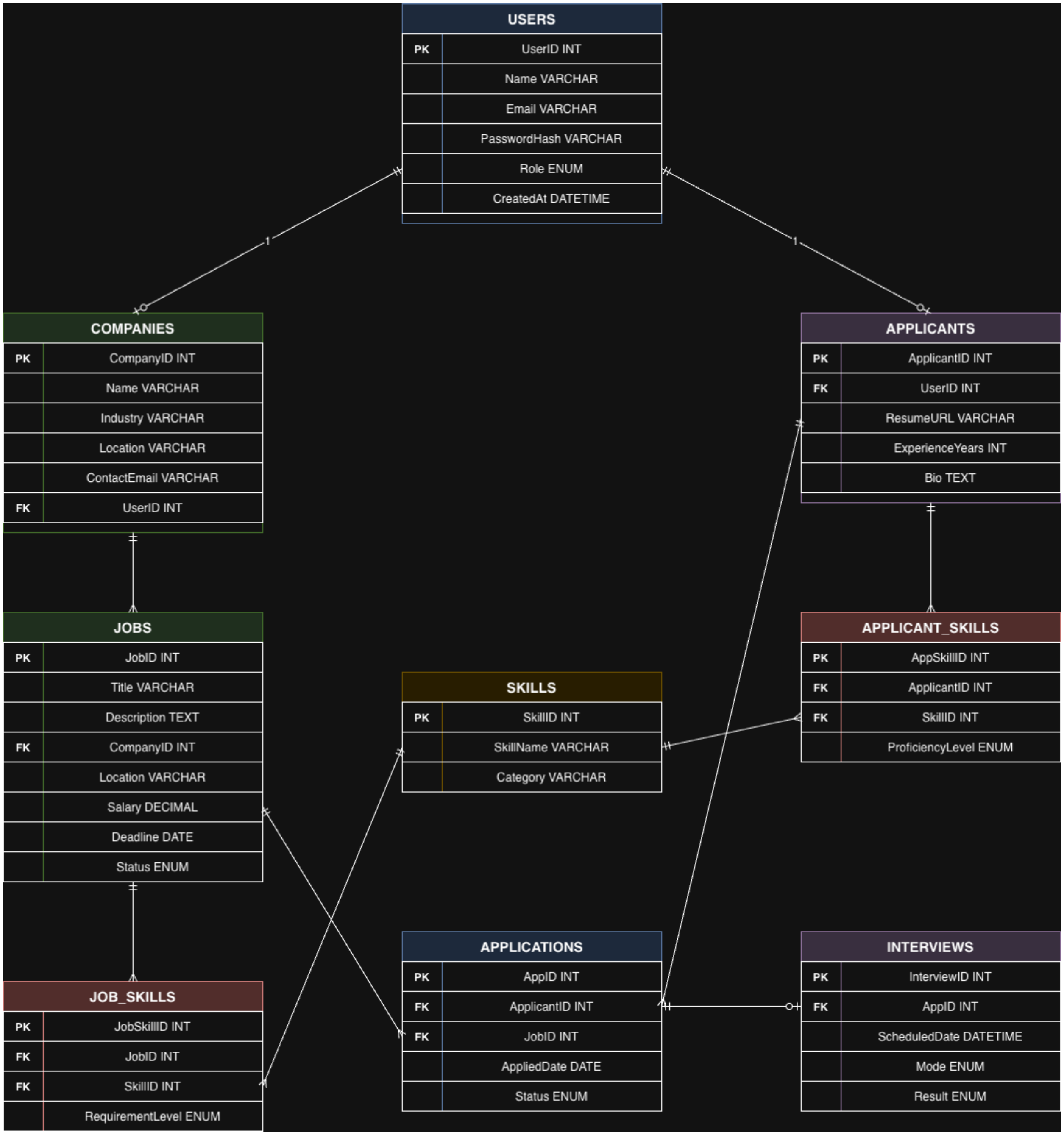
PROJECT MILESTONES

Milestone	Date	Version	Remarks
ERD & Schema Design	April 25, 2026	1.0	—
Database Implementation	—	—	—
Testing & Review	—	—	—



1. Overview

The Job Portal with Intelligent Filtering database is designed as a normalized MySQL relational model that powers a full-stack web application connecting employers with job seekers. The schema prioritizes data integrity, enforcing strict foreign key constraints to manage the complete recruitment lifecycle — from job posting and applicant registration through skill-based matching, application submission, and interview scheduling.

At its core, the system maintains a master skills repository referenced by both job listings and applicant profiles through junction tables, enabling a transparent, query-driven skill-match scoring mechanism. This relational approach eliminates redundancy, ensures referential consistency, and supports the complex filtered queries required for intelligent candidate–job matching.

2. Core Recruitment Management

2.1 Users

This table acts as the central authentication and identity store for all system participants. It manages role-based access control through an ENUM field, distinguishing between Job Seekers, Employers, and Administrators.

Primary Key: UserID

Attributes

- Name, Email — Basic identification and contact information for user profiles.
- PasswordHash — Securely stores hashed credentials; plaintext passwords are never persisted.
- Role (ENUM) — Enforces system-level access boundaries (JobSeeker, Employer, Admin).
- CreatedAt — Timestamp recording account creation for auditing and analytics.

Relationships

- One-to-One (optional) with Companies — an Employer-role user may register a company profile.
- One-to-One (optional) with Applicants — a JobSeeker-role user creates an applicant profile.

2.2 Companies

Defines the organizational entities that post job listings. Every company record is owned by exactly one employer-type user, preventing orphan listings and ensuring accountability.

Primary Key: CompanyID

Attributes

- Name — The registered business name displayed on job postings.
- Industry — Sector classification enabling industry-based filtering and analytics.
- Location — Physical or headquarters location of the organization.
- ContactEmail — Official communication channel for applicant inquiries.
- UserID (FK) — References the employer user who owns this company profile.

Relationship

- One-to-Many with Jobs — a single company can post multiple job listings.

2.3 Jobs

The primary listing entity that stores all details about open positions. Each job belongs to exactly one company and carries structured metadata enabling advanced search and filtering.

Primary Key: JobID

Attributes

- Title — The position name displayed to applicants (e.g., Backend Developer).
- Description (TEXT) — Full-length job description including responsibilities and qualifications.
- CompanyID (FK) — Ensures every listing is associated with a valid, registered company.
- Location — Work location for the position, enabling geographic filtering.
- Salary (DECIMAL) — Offered compensation, supporting salary-range queries.
- Deadline (DATE) — Application closing date; expired listings can be filtered automatically.
- Status (ENUM) — Lifecycle state of the listing (Open, Closed, Draft).

Relationships

- Many-to-One with Companies — each job belongs to one company.
- One-to-Many with Job_Skills — a job can require multiple skills.
- One-to-Many with Applications — a job receives multiple applications.

2.4 Applicants

Extends the base user record with job-seeker-specific profile data. This separation follows normalization principles — only users with the JobSeeker role require these additional attributes.

Primary Key: ApplicantID

Attributes

- UserID (FK) — Links the applicant profile to its parent user account.
- ResumeURL (VARCHAR) — Path or link to the uploaded resume document.
- ExperienceYears (INT) — Total professional experience, used in experience-based filtering.
- Bio (TEXT) — Free-text professional summary for employer review.

Relationships

- One-to-Many with Applicant_Skills — an applicant can possess multiple skills.
- One-to-Many with Applications — an applicant can submit multiple applications.

3. Skill-Based Matching Engine

The intelligent filtering capability of the system is powered by a normalized skill-matching architecture. Rather than storing skills as unstructured text, the schema maintains a centralized Skills master table referenced by two junction tables — one linking skills to jobs and another linking skills to applicants. This design enables the system to compute a transparent skill-match percentage through standard SQL queries.

3.1 Skills (Master Table)

A centralized, canonical repository of all professional skills recognized by the platform. This master list prevents duplication and ensures consistent skill references across both job requirements and applicant profiles.

Primary Key: SkillID

Attributes

- SkillName (VARCHAR) — The unique, standardized name of the skill (e.g., Python, SQL, Project Management).
- Category (VARCHAR) — Logical grouping for filtering (e.g., Programming, Data Science, Soft Skills).

Purpose

Acts as the single source of truth for skill definitions. Both Job_Skills and Applicant_Skills reference this table, ensuring that comparisons between job requirements and applicant qualifications are made against a consistent taxonomy.

3.2 Job_Skills (Junction Table)

Resolves the many-to-many relationship between Jobs and Skills by recording which skills each job requires, along with the expected proficiency level.

Primary Key: JobSkillID

Foreign Keys: JobID, SkillID

Attributes

- RequirementLevel (ENUM) — The minimum proficiency expected: Beginner, Intermediate, or Expert.

Relationship

Many-to-Many between Jobs and Skills — a single job can require multiple skills, and a single skill can appear in the requirements of many jobs.

3.3 Applicant_Skills (Junction Table)

Resolves the many-to-many relationship between Applicants and Skills by recording which skills each applicant possesses, along with their self-assessed proficiency.

Primary Key: AppSkillID

Foreign Keys: ApplicantID, SkillID

Attributes

- ProficiencyLevel (ENUM) — The applicant’s self-declared competency: Beginner, Intermediate, or Expert.

Relationship

Many-to-Many between Applicants and Skills — an applicant can possess many skills, and a single skill can be claimed by many applicants.

4. Application Lifecycle & Status Tracking

The application subsystem manages the complete hiring pipeline from initial submission through final disposition. It provides real-time status visibility for applicants and structured candidate management for employers.

4.1 Applications

The transactional core of the recruitment process. Each record represents a single applicant's interest in a specific job, tracking the application through a defined status pipeline.

Primary Key: AppID

Attributes

- ApplicantID (FK) — The applicant who submitted the application.
- JobID (FK) — The job listing being applied to.
- AppliedDate (DATE) — Timestamp of submission, enabling recency-based sorting.
- Status (ENUM) — Current pipeline stage: Applied, Shortlisted, Rejected, or Hired.

Constraints

- UNIQUE (ApplicantID, JobID) — Prevents duplicate applications by the same applicant to the same job.

Relationships

- Many-to-One with Applicants — an applicant submits many applications.
- Many-to-One with Jobs — a job receives many applications.
- One-to-One (optional) with Interviews — an application may have one scheduled interview.

4.2 Interviews

Extends the application lifecycle by recording scheduled interviews linked to specific applications. This table supports the employer's ability to move candidates from shortlisting into the interview stage.

Primary Key: InterviewID

Attributes

- AppID (FK) — The application for which this interview is scheduled.
- ScheduledDate (DATETIME) — The planned date and time of the interview.
- Mode (ENUM) — Interview format: Online or Onsite.
- Result (ENUM) — Outcome once conducted: Pending, Passed, or Failed.

Relationship

One-to-One with Applications — each application can have at most one interview record, ensuring a clean audit of the interview stage.

5. Data Integrity Constraints

The schema enforces data quality and consistency at the database level through a comprehensive set of constraints, eliminating reliance on application-layer validation alone.

Foreign Key Constraints

- Every Application references a valid ApplicantID and JobID, preventing orphan records.
- Every Job references a valid CompanyID; every Company references a valid UserID.
- Junction tables (Job_Skills, Applicant_Skills) enforce valid references to both parent entities.
- Every Interview references a valid AppID, ensuring interviews cannot exist without an application.

UNIQUE Constraints

- UNIQUE (ApplicantID, JobID) on Applications prevents an applicant from submitting duplicate applications to the same job.
- UNIQUE (Email) on Users prevents duplicate account registration.

NOT NULL Constraints

- All primary keys and critical identification fields (Name, Email, Title) are marked NOT NULL.
- Foreign key columns are NOT NULL where the relationship is mandatory (e.g., a Job must always reference a CompanyID).

ENUM Constraints

- Users.Role restricts values to JobSeeker, Employer, Admin.
- Jobs.Status restricts values to Open, Closed, Draft.
- Applications.Status restricts values to Applied, Shortlisted, Rejected, Hired.
- Interviews.Mode restricts values to Online, Onsite.
- Interviews.Result restricts values to Pending, Passed, Failed.
- RequirementLevel and ProficiencyLevel restrict values to Beginner, Intermediate, Expert.

6. Reporting & Analytics

The normalized schema is designed to support a variety of operational and strategic reports through standard SQL aggregation and join queries.

Supported Reports

- Applications per Job — Aggregate count of applications received for each listing, with status breakdown.
- Hiring Funnel Analysis — Conversion rates across pipeline stages (Applied → Shortlisted → Hired) per company or job.
- Skill Demand Trends — Frequency analysis of skills across all active job listings, identifying in-demand competencies.
- Applicant Match Scoring — Percentage of required skills matched per applicant per job, ranked by relevance.
- Employer Dashboard Metrics — Total listings, active positions, pending applications, and interview schedules per company.

7. Relational Schema Summary

The following table summarizes all entities in the Job Portal schema, their primary keys, foreign key dependencies, and inter-table relationships.

Entity	PK	Foreign Keys	Type	Key Relationship
Users	UserID	—	Base	Parent of Companies, Applicants
Companies	CompanyID	UserID	Base	1:N with Jobs
Jobs	JobID	CompanyID	Base	1:N with Job_Skills, Applications
Applicants	ApplicantID	UserID	Base	1:N with Applicant_Skills, Applications
Skills	SkillID	—	Master	Referenced by Job_Skills, Applicant_Skills
Job_Skills	JobSkillID	JobID, SkillID	Junction	M:N between Jobs and Skills
Applicant_Skills	AppSkillID	ApplicantID, SkillID	Junction	M:N between Applicants and Skills
Applications	AppID	ApplicantID, JobID	Transaction	1:1 (opt.) with Interviews
Interviews	InterviewID	AppID	Extension	1:1 with Applications

8. Summary of System Design

The Job Portal with Intelligent Filtering database schema comprises 9 interrelated entities organized into three functional layers:

- Core Recruitment Management — Users, Companies, Jobs, and Applicants form the foundational identity and listing layer, managing authentication, organizational profiles, job postings, and applicant records.

- Skill-Based Matching Engine — The Skills master table, together with the Job_Skills and Applicant_Skills junction tables, powers the intelligent filtering system by enabling transparent, query-driven skill-match scoring between candidates and positions.
- Application Lifecycle — The Applications and Interviews tables manage the recruitment pipeline from submission through interview scheduling and final disposition, with ENUM-constrained status fields ensuring valid state transitions.

The schema is fully normalized to Third Normal Form (3NF), eliminating data redundancy while maintaining referential integrity through foreign key constraints, UNIQUE constraints on critical pairs, NOT NULL enforcement on mandatory fields, and ENUM restrictions on status columns. This design supports the full spectrum of CRUD operations, complex filtered queries, and aggregated reporting required by a production recruitment platform, while remaining practical in scope for a semester-level database systems project.